

# IBD Patient face higher Incidence of Anal Dysplasia and Cancer, Hightened by HPV, Perianal Disease

*Introduction:* Patients with inflammatory bowel disease (IBD) have increased rates of anal cancer. We evaluated whether high-risk HPV-positive results from anal sampling or perianal disease was associated with higher anal cancer or dysplasia prevalence in IBD.

*Methods:* We performed a retrospective analysis of Epic Cosmos, an anonymized database of 305,065,760 patients. Patients with active diagnoses from 6/1990 to 6/2026 were stratified as Crohn's disease, ulcerative colitis, or non-IBD and compared for anal cancer/dysplasia, anal HR-HPV+ DNA, and perianal disease.

Anal HR-HPV+ status, anal cancer/dysplasia, and perianal disease were defined by ICD-10 codes. Anal cancer/dysplasia included HGIL of anus, anal intraepithelial neoplasia III, or dysplasia of anus. Perianal disease included perianal fistula, abscess, or fissure. Chi-square testing used  $p < 0.05$ . Although the source population included approximately 2.4 million IBD patients, testing denominators, negative results, screening frequency, and HPV vaccination status were unavailable; findings may reflect screening, surveillance, or coding differences.

*Results:*

| Patient Type                        | Anal Cancer or Dysplasia (total) | Anal Cancer or Dysplasia 5 years after Perianal Disease diagnosis | Anal cancer or dysplasia 5 years after HR-HPV diagnosis | Positive High-Risk HPV diagnosis in anal area (total) |
|-------------------------------------|----------------------------------|-------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------|
| Non-IBD<br>(n=305,065,760)          | 128,173<br>(0.0420%)*            | 10,195<br>(0.0033%)*                                              | 9,404<br>(0.0031%)*                                     | 22,867<br>(0.0075%)*                                  |
| Crohn's Disease<br>(n=1,206,873)    | 2,535<br>(0.2100%)*              | 575 (0.0476%)*                                                    | 97 (0.0080%)*                                           | 197 (0.0163%)*                                        |
| Ulcerative Colitis<br>(n=1,191,972) | 3,227<br>(0.2707%)*              | 488 (0.0409%)*                                                    | 149 (0.0125%)*                                          | 319 (0.0268%)*                                        |

\*=Row-level Pearson chi-square test  $p < 0.001$ .

Compared with non-IBD patients, anal cancer/dysplasia was 5.0-fold higher in Crohn's disease and 6.4-fold higher in ulcerative colitis. Anal HR-HPV positivity was 2.2-fold higher in Crohn's disease and 3.6-fold higher in ulcerative colitis. Anal cancer/dysplasia within 5 years after anal HR-HPV diagnosis was also higher in IBD, with a 2.6-fold increase in Crohn's disease and 4.1-fold increase in ulcerative colitis. The largest relative increase followed a perianal disease diagnosis, with anal cancer/dysplasia within 5 years 14.3-fold higher in Crohn's disease and 12.3-fold higher in ulcerative colitis. Unadjusted odds ratios versus non-IBD were similar to

observed frequency ratios, with the largest associations seen after perianal disease diagnosis: OR 14.26 for Crohn's disease and OR 12.26 for ulcerative colitis.

*Conclusion:* Patients with IBD had higher observed frequencies of anal cancer/dysplasia than non-IBD patients. There were higher frequencies of cancer/dysplasia diagnosis within 5 years after an anal-area high-risk HPV or perianal disease diagnosis. The strongest association was seen in Crohn's disease after perianal disease diagnosis. These findings support further study of targeted anal cancer screening strategies in IBD patients with anal HR-HPV positivity or prior perianal disease.